

The two-piece $\mathfrak{B}_3$ row

Checking the p^+ integration term by term

$$A \equiv (x-z)^2, \quad B \equiv (x-w)^2, \quad \beta \equiv \frac{k^+ B}{A}, \quad \text{units below: the common } \frac{ig^4 f^{abd}}{16\pi^5} = \frac{ig^4 f^{abd}}{8\pi^4} \cdot \frac{1}{2\pi}$$

The algebra is correct. Both prefactors reproduce with ratio exactly 1, and all three p^+ integrals agree with principal-value quadrature to 10^{-8} – 10^{-12} across four configurations (§2). Every logarithm combines the way you wrote it — in particular your blue $\left[\log \frac{\vee}{\Lambda} + \log(1 - \frac{k^+}{\vee})\right]$ is exactly $\log \frac{\vee - k^+}{\Lambda}$, and $\left[\log \frac{\vee}{k^+ - \Lambda} - \log \frac{\vee}{\Lambda}\right]$ is exactly $-\log \frac{k^+ - \Lambda}{\Lambda}$ (§3). **But the result is incomplete in one physical respect:** it is the principal value, and the on-shell $i\pi$ piece is missing (§5) — which matters here because every prefactor carries an explicit i , so that piece is **real**. And the $\frac{1}{\Lambda}$ is a **power** divergence with nothing left to cancel it now that piece II is gone (§4).

1. The input expression

piece	quoted coefficient	computed / quoted
I (the δ_{ij} one, terms 1–2)	$\frac{ig^4 f^{abd}}{8\pi^4} \frac{(p^+ + k^+)}{(k^+ - p^+)^2}$	1
III (term 3)	$-\frac{ig^4 f^{abd}}{8\pi^4} \frac{1}{p^+ k^+}$	1

The argument map $\mathfrak{B}_3(k^+, z; p^+, -u) \rightarrow \mathfrak{B}_3(k^+, w; p^+, -z)$ sends $(x-z)_{\text{def}} \rightarrow (x-w)$ and $(x-u)_{\text{def}} \rightarrow (x-z)$, so $p^+(x-u)^2 - k^+(x-z)^2 \rightarrow p^+A - k^+B$ as you have it. The colour $f^{abd}[U^{de}(x)U^{bc}(z) - U^{de}(x)U^{bc}(y)]\rho^e(x)\rho^c(y)$ is the structure checked before, with f^{abd} legitimately an overall factor.

Two carry-overs. Term 3 of the *pre*-integration expression has no $\int_{x,y,z,x'}$ (the post-integration terms all do). And the bra bracket still has the $U^{ab'}(w') / U^{b'a}(w')$ transpose — the row gives the latter.

2. The three p^+ integrals, checked

With $D = p^+A - k^+B = A(p^+ - \beta)$, the primitives are

$$F_I(p^+) = -\frac{2}{(A-B)(p^+ - k^+)} + \frac{A+B}{k^+(A-B)^2} \log \left| \frac{p^+ - \beta}{p^+ - k^+} \right|, \quad F_{\text{III}}(p^+) = \frac{1}{k^+} \left[-\log p^+ + 2 \log |p^+ - \beta| \right],$$

both exact ($\frac{dF}{dp^+} - \text{integrand} = 0$ symbolically). Evaluating them on your three ranges and comparing with direct PV quadrature:

$(A, B, \Lambda, \vee)$	piece	PV quadrature	your result	rel. diff
(0.62, 1.45, 10^{-5} , 60)	I on $[k^+ + \Lambda, \vee]$	−240999.35396372	−240999.35396412	1.6×10^{-12}
	I on $[\Lambda, k^+ - \Lambda]$	−240928.52818060	−240928.52817951	4.6×10^{-12}
	III on $[\Lambda, \vee]$	9.19728884	9.19728805	8.6×10^{-8}
(1.80, 0.55, 10^{-6} , 120)	I	1599979.76533078	1599979.76519915	8.2×10^{-11}
	I	1600020.41323624	1600020.41328228	2.9×10^{-11}
	III	6.66186405	6.66186403	3.5×10^{-9}
(2.40, 1.10, 10^{-6} , 300)	I	1538434.19492623	1538434.19479966	8.2×10^{-11}
	I	1538488.95792701	1538488.95797120	2.9×10^{-11}
	III	6.55446442	6.55446450	1.1×10^{-8}

(A fourth configuration, (0.90, 2.07, 10^{-5} , 40), agrees equally well.) The residual differences are the quadrature's, not the algebra's.

3. Why the logarithms combine the way you wrote them

The log part of F_I on $[k^+ + \Lambda, \vee]$ is

$$\frac{A+B}{k^+(A-B)^2} \left[\log \frac{|\vee A - k^+ B|}{|(k^+ + \Lambda)A - k^+ B|} - \log \frac{\vee - k^+}{\Lambda} \right],$$

and your two terms supply exactly that, because

$$-\log \frac{\vee}{\Lambda} - \log \left(1 - \frac{k^+}{\vee} \right) = -\log \frac{\vee}{\Lambda} - \log \frac{\vee - k^+}{\vee} = -\log \frac{\vee - k^+}{\Lambda} \quad \checkmark$$

Similarly on $[\Lambda, k^+ - \Lambda]$ the primitive gives $\log \frac{k^+ - \Lambda}{\Lambda} + \log \frac{|(k^+ - \Lambda)A - k^+ B|}{|\Lambda A - k^+ B|}$, and your second blue bracket is

$$-\left[\log \frac{\vee}{k^+ - \Lambda} - \log \frac{\vee}{\Lambda} \right] = \log \frac{k^+ - \Lambda}{\Lambda} \quad \checkmark$$

— the $\vee$ cancels inside it, as it should. And piece III gives $\frac{1}{k^+} \left[-\log \frac{\vee}{\Lambda} + 2 \log \left| \frac{\vee A - k^+ B}{\Lambda A - k^+ B} \right| \right]$, matching your last two terms once you note $\frac{k^+ B - \vee A}{k^+ B - \Lambda A} = \frac{\vee A - k^+ B}{\Lambda A - k^+ B}$.

4. The $\frac{1}{\Lambda}$ is a power divergence, and now nothing cancels it

Your terms 1 and 4 both carry $+\frac{1}{\Lambda}$ with the **same** coefficient $\frac{1}{A-B}$, so they add rather than cancel:

$$(\text{terms 1 and 4}) \longrightarrow \frac{4}{\Lambda(A-B)} \quad \text{in the units above.}$$

(A, B, Λ)	pieces 1+2, PV	$\frac{4}{\Lambda(A-B)}$
(0.62, 1.45, 10^{-5})	-481927.88	-481927.71
(1.80, 0.55, 10^{-6})	3200000.18	3200000.00
(2.40, 1.10, 10^{-6})	3076923.15	3076923.08

In the three-piece version of $\mathfrak{B}_3$ the two halves of piece II had equal and opposite double poles at $p^+ = k^+$ and cancelled each other, but piece I's was *still* uncanceled. Here, with piece II absent altogether, **there is nothing to cancel it at all**. And $\frac{1}{\Lambda}$ is not a rapidity logarithm: evolution cannot absorb it. Either it cancels against a companion row, or the Θ regularisation of the $p^+ = k^+$ pole needs rethinking.

5. What the result is missing: the on-shell piece

$p^+ A - k^+ B$ vanishes at $\beta = k^+ \frac{(x-w)^2}{(x-z)^2}$, and for generic transverse points that sits **inside** the integration range:

A	B	β	in $[\Lambda, k^+ - \Lambda]$?	in $[k^+ + \Lambda, \vee]$?	in $[\Lambda, \vee]$?
0.62	1.45	2.3387	no	yes	yes
1.80	0.55	0.3056	yes	no	yes
2.40	1.10	0.4583	yes	no	yes

So the integral is only defined once you supply a prescription. It is a light-cone energy denominator, so the propagator's $+i\epsilon$ gives

$$\frac{1}{p^+ A - k^+ B \pm i\epsilon} = \text{PV} \frac{1}{p^+ A - k^+ B} \mp i\pi \delta(p^+ A - k^+ B).$$

What you wrote is the principal value. The missing δ pieces are, in the same units,

$$\text{pieces I : } \mp i\pi \frac{A+B}{k^+(A-B)^2} \quad (\text{whenever } \beta \text{ lies in that range}), \quad \text{piece III : } \pm \frac{2i\pi}{k^+}.$$

Note the first is *the same coefficient that multiplies the logarithm* — as it must be, since both come from the residue at $p^+ = \beta$.

These are not optional. Every prefactor in this row carries an explicit i , so $i \times (\mp i\pi)$ is **real** and contributes directly to the cross section. Physically it is the on-shell intermediate state — a genuine real-emission contribution. Dropping it loses a real part of the answer.

5.1 A related bookkeeping point

Because β can lie on either side, the arguments of your logarithms change sign. For $A = 0.62$, $B = 1.45$, $\Lambda = 10^{-5}$:

$$(\Lambda + k^+)A - k^+B = -0.8300, \quad \vee A - k^+B = +35.7500,$$

so the ratio inside $\log \left[\frac{\vee A - k^+B}{(\Lambda + k^+)A - k^+B} \right]$ is **negative**. The expressions only make sense as $\log |\cdot|$ — which is exactly what the principal value produces, and what the numerical agreement in §2 confirms. Write the absolute-value bars explicitly, so the branch is not left to chance when the imaginary parts are restored.

6. Summary

item	status
prefactors of both $\mathfrak{B}_3$ pieces	correct, ratio exactly 1
argument map and colour structure	correct
all three p^+ integrals	correct as principal values, to 10^{-8} – 10^{-12}
the two blue logarithm combinations	correct ($= \log \frac{\vee - k^+}{\Lambda}$ and $\log \frac{k^+ - \Lambda}{\Lambda}$)
the $\frac{1}{\Lambda}$ terms	correct, but they add to $\frac{4}{\Lambda(A-B)}$ — a power divergence with nothing to cancel it
on-shell $i\pi$ pieces	missing — and they are real here
$\log$ vs $\log \cdot $	write the bars
measure of pre-integration term 3; bra transpose	still to fix

zint/src/row16_check.py (§2–3), row16_caveats.py (§1, §4–5) — every number above.